Counting, Part 2

Session II · 14 September 2026 — case definitions, counted deaths, the pyramid of lost cases

Class notes · an automated summary of the live session, structured for study. Where these notes and the assigned papers disagree, the papers win.

Key outcomes

Part two of the counting session. The central thread: what gets counted depends on how cases are defined, what tests exist, and what reporting incentives are at work — so observed statistics systematically underrepresent the true burden. The session walked from case definitions (HIV tests and their blind window, long COVID's four competing definitions) through drug-death classification, to the pyramid of lost cases between true infection and national statistics, and the ways test availability alone can manufacture or hide an epidemic.

Case definition: what makes someone a case

HIV

- A case = a person testing positive on a recognised diagnostic; two main test families:
  - PCR — detects the virus's nucleic acids directly; works early, costs more
  - ELISA — detects antibodies; cheaper, ubiquitous, but needs the immune response to develop
- The blind window: right after exposure antibodies do not yet exist — ELISA returns a false negative while PCR could already see the infection; routine screening often funds only ELISA
- A full case definition also needs clinical criteria, an exposure window, and reportable status that obliges the clinician to notify the system

Long COVID: a case-definition problem in real time

- At least four definitions in simultaneous use (WHO's three-month rule, NICE's split into ongoing symptomatic and post-COVID, and others) — producing incomparable statistics across countries
- Criteria lean on self-report, so recall bias, stigma and reporting culture enter the count directly
- Confounders are everywhere: financial stress, family crisis, or another infection in the window produce the same fatigue and low mood the criteria accept
- The diagnosis remains contested — and is still used in places to justify procedures and treatment for post-infectious asthenia

Counting the dead

- A drug-related death needs an autopsy, a toxicology panel (someone chooses what to test for), and an intent classification under ICD: accidental, intentional self-poisoning, or undetermined
- Classification is done by a pathologist under institutional practices — so statistics differ systematically between hospitals, cities and countries
- Polydrug reality: in 2016 about 80% of synthetic-opioid deaths in the USA involved other substances or alcohol; in 2017 about 70% of cocaine deaths involved an opioid
- Fentanyl enters other drugs because it is cheap and potent; combined with alcohol or benzodiazepines it multiplies respiratory depression. It is less a party drug than a cheap substitute
- Harm reduction in one line each: syringe exchange cuts bloodborne transmission, not drug use; methadone improves survival and life quality, not the epidemic; alcohol — the most burdensome addiction — is legitimised and has the fewest programmes

The pyramid of lost cases

- Between true infection and the reported number, every step drops a fraction: susceptible → infected → symptomatic → seeking care → tested → test-positive → reported
- Reported cases = true incidence × a chain of multipliers, each below one; the true number is always higher
- Example scale: ~98% of poliomyelitis infections are asymptomatic — the visible disease is the tip
- Administrative pressure adds its own filter: facilities can avoid registering dangerous diagnoses to escape quarantine, penalties or reputational damage

Endemic vs epidemic

- Endemic = a stable local reservoir sustains transmission (malaria where its Anopheles host breeds); a travel-imported case in London does not make Ebola endemic there — no local reservoir
- The distinction matters for reading surveillance: imported cases and local transmission are different objects

When the test system writes the curve

- Lyme disease (USA): reported cases roughly doubled between 2017 and 2022 after antibody testing became ubiquitous — part real, part over-attribution of chronic fatigue in the absence of alternatives
- Hepatitis C (USA): rising counts reflect a real epidemic riding the opioid crisis — distinguishable from the Lyme pattern because mortality and transmission routes move together with it
- Thyroid cancer (Korea): mass ultrasound screening multiplied detected cases while death rates stayed flat — an apparent epidemic of clinically insignificant disease
- COVID-19 under zero-tolerance policy: any positive PCR — including false positives — triggered restrictions, inflating counts and distorting policy

Field reality: an infectious diseases hospital

- The instructor's background: Botkin Hospital, St Petersburg — wards for AIDS, tuberculosis, intestinal infections, tropical disease
- A ward for incarcerated and formerly incarcerated patients treated HIV co-infected with drug-resistant tuberculosis — resistance grown by interrupted therapy in prisons, often disseminated and aggressive
- A diagnostic holding ward admitted patients without a confirmed diagnosis — where test systems were missing, pending, or nonexistent for rare tropical infections
- High hepatitis B/C and HIV burdens in the Caucasus region; programmes exist, but test systems and treatment access are uneven

Self-report under stigma

- Questionnaires about stigmatised behaviour (drugs, alcohol, sex) underestimate systematically — the bias is directional, always downwards
- It measures willingness to report, not the behaviour
- Partial fix: supplement self-report with random objective checks; or at minimum, state the direction of the bias in the conclusions
- The same mechanism corrupts historical HIV baselines: under stigma, many never tested at all

Deferred to future sessions

- False positives and negatives, sensitivity and specificity, and their arithmetic
- Selection and information bias as formal concepts

- How meta-analysis behaves when it pools studies with different case definitions

Check before you lean on it

- The name of the diagnostic holding ward did not survive transcription; the function described (admission without confirmed diagnosis) is what matters.
- The 2016/2017 polydrug percentages are as spoken in class; verify against CDC sources before citing in written work.